Mohammed Abdur Rehman

ai23btech11015@iith.ac.in | abdurrehman0105@gmail.com | +91 9059473549 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.35
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2023	97.70%
Narayana concept School	Class X (Board of Secondary Education)	2021	10.00%

EXPERIENCE

AI Engineer Intern - Dunkit Inc.

- Architected an end-to-end AI surveillance and LLM pipeline enabling natural language querying of video data.
- Engineered real-time tracking using YOLOv8 and custom Euclidean centroid algorithms to maintain persistent target IDs.
- Implemented spatial anomaly detection to flag entry/exit events and suspicious target proximity across frames.
- Designed event-driven video archiving utilizing circular frame buffers to automatically extract full incident clips.
- Developed an intelligent search interface via FastAPI and local LLMs (Ollama/Mistral) for real-time video retrieval.

AI/ML Developer - LV Prasad Eye Institute (Grabi App),

- Engineered a 2-stage deep learning pipeline converting smartphone eye photos to clinical slit-lamp quality for tele-ophthalmology.
- Synthesized 1,700+ paired training images via domain adaptation using a VGG19 Neural Style Transfer framework.
- A Mask-Guided Restormer Transformer to execute targeted, high-fidelity image enhancement on low-quality inputs.
- Integrated an EfficientNet-B4 U-Net++ segmentation mask to accurately preserve critical anatomical structures.
- Evaluated models by fine-tuning transformer architectures and building comprehensive benchmarking pipelines for quality assessment

PROJECTS

MASooda: Hierarchical, Priority-Aware, Resilient Urban Air Traffic Management

- Built the simulation engine for 60 heterogeneous agents: double-integrator dynamics, k-NN/radius communication graph, and 3-corridor environment.
- Led end-to-end integration of ADMM conflict resolution, boids behavior, global overseer, and W-MSR into a one-command pipeline with fallbacks.
- Implemented live Signal Temporal Logic (STL) runtime monitoring of minimum-separation safety, logging robustness, violations, and recovery time.
- Benchmarked hierarchical vs. centralized vs. decentralized-undefended architectures under identical seeds, with a centralized baseline and common metrics pipeline.
- Developed a real-time dashboard and reproducible experiment scripts for the final demo.

Camera Surfers - Real-Time Webcam Pose Controller

- Developed a real-time gesture-control desktop application that translates physical body movements (jumping, crouching, lane stepping) into low-latency keyboard inputs for *Subway Surfers* without modifying game binaries.
- Implemented **MediaPipe 3D pose tracking** ; engineered dynamic torso-scale normalization, rolling percentile baseline filters, and velocity gating to accurately classify jumps, crouches, and lane shifts

DocuMind - Multimodal Document Intelligence Agent

- Built an agentic multimodal RAG system that reasons across text, tables, scanned pages, and visual document content for financial and legal analysis.
- Designed page-level visual retrieval and structured table extraction to let the agent trace answers back to their exact document evidence.

- Implemented multi-hop verification to cross-check numerical claims across narrative text, tables, and different sections of a document.
- Achieved **88% Retrieval Recall@5, 82% Citation Accuracy, and 3% hallucination rate** through an automated evaluation pipeline with grounded Q&A testing.

Agentic Legal RAG Mesh over Indian Jurisprudence

- Built a modular multi-agent workflow using LangGraph to parallelize statutory lookup, precedent retrieval, and draft generation.
- Implemented hybrid BM25 and Qdrant dense vector search over central statutes, boosting citation precision by **+25%** over single-prompt RAG.
- Created a GPT-4o-mini LLM-as-a-judge critic pipeline with conditional graph routing to eliminate hallucinations and verify legal references.
- Developed a Redis-backed FastAPI engine featuring real-time SSE streaming and Streamlit UI, fully containerized using Docker Compose.

Adaptive Learning and Path-Planning Engine

- Built a personalized learning engine with a curriculum prerequisite graph, adaptive diagnosis and planner, and evolving mastery model updated from quizes.
- Developed a study planner that adjusts to the learner's time and deadline, adds reviews and remediation, and replans when sessions are missed, compared against simple baselines.

Indian Handicrafts Portal - NLP Course Project

- Built a Next.js handicraft discovery portal with searchable, filterable, and state-based exploration across 200+ Indian handicrafts.
- Developed a similarity-based duplicate detection workflow and prototype NLP capabilities using TF-IDF search, MiniLM embeddings, and extractive text summarization.
- Created an interactive chatbot, India craft map, detailed craft pages, and community registration workflow with browserbased persistence.

Frequency-Domain Visual Prompting for Test-Time Adaptation (Computer Vision / Prof. M Konda Reddy)

- Developed a frequency-domain TTA method using FFT, radial frequency features, and an MLP to generate input-dependent visual prompts.
- Adapted image representations by modifying frequency amplitudes while preserving phase, with entropy minimization and L2 regularization for stable test-time updates.
- Evaluated the method on CIFAR-10-C and PACS using a frozen ResNet-18, analyzing performance under corruption and semantic domain shifts.

SKILLS

Languages: Python,C, C++,SQL

Libraries: NumPy, Pandas, Matplotlib, Scikit-learn,Pytorch

Tools: Git, LaTeX, Linux

Familar with: HTML, CSS, JavaScript,React,Node.js, Express.js,TensorFlow

RELEVANT COURSES

AI: Foundations of Machine Learning, Deep Learning, Online Learning,Computer Vision, Natural Language Processing ,Distributed in AI Training

CS: Computer Architecture, Discrete Mathematics, DBMS, Data Structures and Applications, Algorithms, Introduction to Programming,

Mathematics: Probability and Random Variables, Matrix Theory,Calculus, Vector Calculus, Differential Equations, Applied Statistics

EXTRA CURRICULAR ACTIVITIES

- National Services Scheme(NSS) - IITH - Student Volunteer(86 hours)
- I like to play, watch and analyse cricket
- Helping students with EAMCET and IPE exams
- I played competitive BGMI and Clash Royale at college level

POSITIONS OF RESPONSIBILITIES

- Esports Coordinator at Glitch Club
- Worked as Transport coordinator for workshops